Systems Engineering - Midterm Review

Team 9 - Semantic Transmitter - "Semantic Audio Communication using GenAI"

André Jacinto, João Abade, Carlos Sousa, José Santos, Eduardo Medeiros, Nuno Vieira, Guilherme Ferreira, Rodrigo Resende, Inês Faleiro, Teresinha Teixeira, Tomás Sousa

Faculdade de Engenharia - Universidade do Porto

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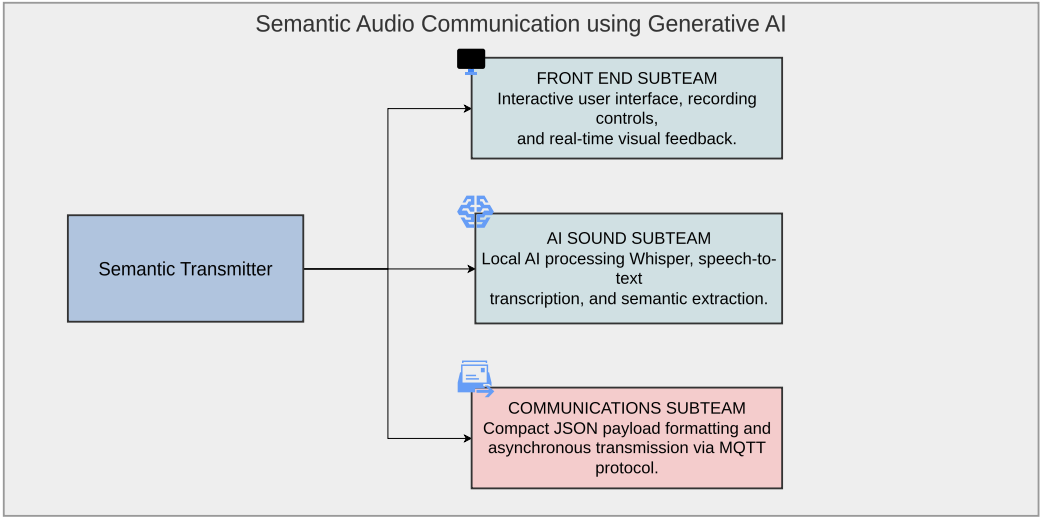
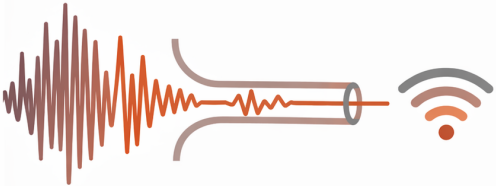
Team organization



Challenge Description

The Problem (Status Quo)



Header: Traditional Voice Messaging

* High bandwidth consumption.
* Severe delivery delays on unstable links.

The Challenge/Our Solution

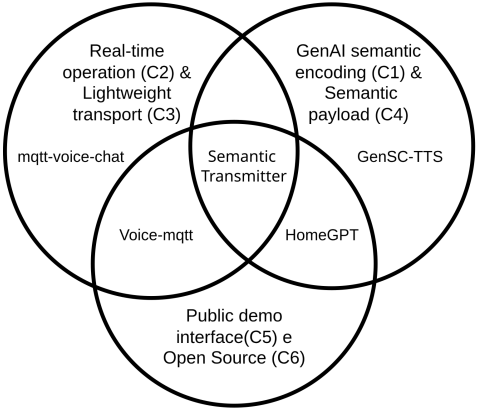
Header: Semantic Transmission

* Edge AI local meaning extraction.
* Smaller payload containing semantic information.
* Capable of working in unstable network conditions.

Validated Requirements

Category	Validated requirements
Communication	Asynchronous message sending
	Standardized semantic payload with transcript and metadata
	Delivery-state feedback
	Measurable bandwidth reduction vs. raw audio transmission
User Interface	Explicit start/stop recording controls
	Dynamic transcript display after processing
	Real-time status feedback
	Graceful, non-disruptive error handling during demos
AI STT Model	Model runs locally; retry path for capture/transcription failures; transcript shown before transmission; raw audio discarded after transcription; low end-to-end latency
Validation concept	Peer Reviews & Checklist Comparison

Outcome of the Market Survey



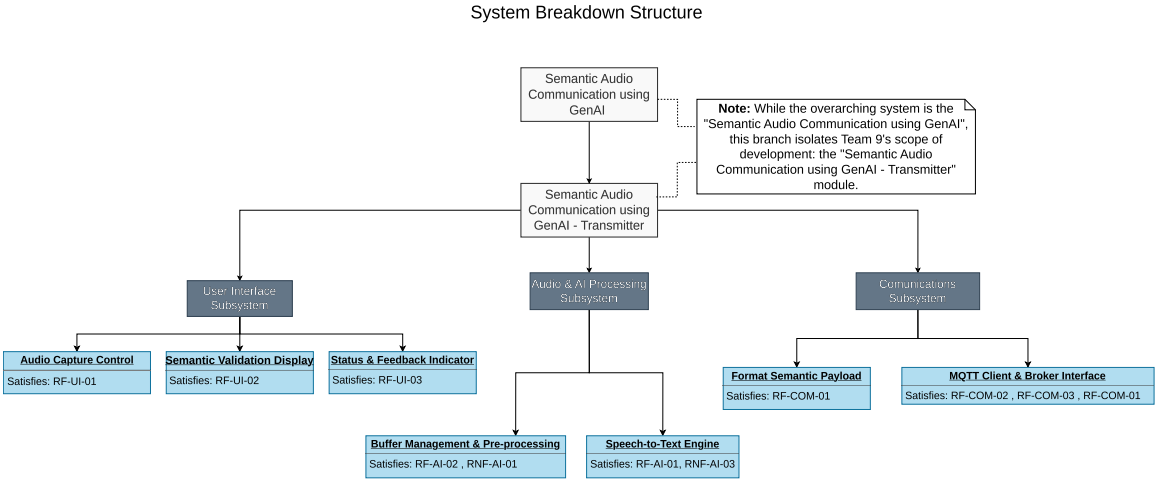
AI PROCESSING: Whisper large-v3

- High Accuracy STT & Native Portuguese Support.
- 100% Offline Capable (Critical for live demos)
- Free & Open-Source

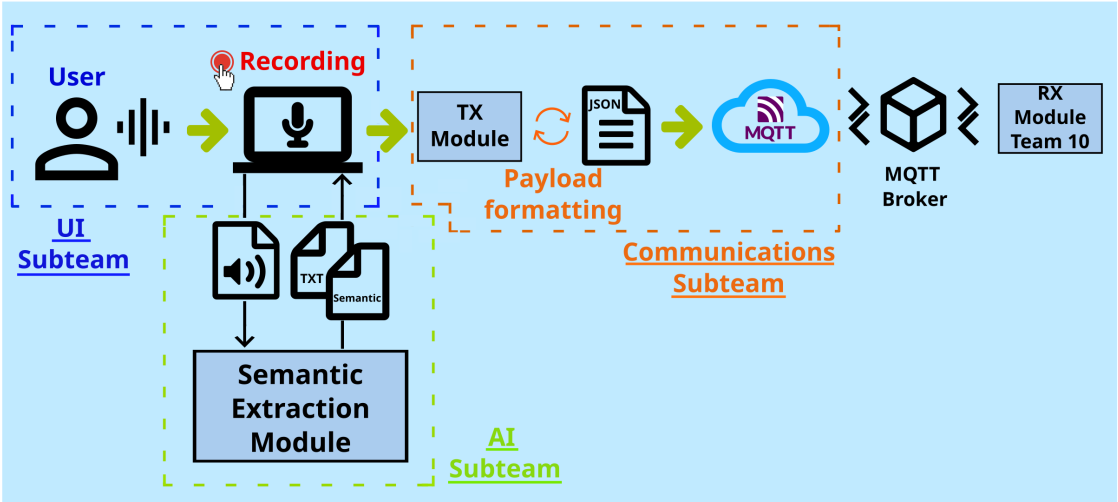
COMMUNICATION TRANSPORT: MQTT

- Minimal network overhead & very low latency
- Pub/Sub Broker Paradigm
- Asynchronous delivery (Decouples TX from RX)

System Breakdown Structure



Functional Architecture



Measures of Effectiveness and Performance

WER

Semantic Accuracy

- Evaluates the system's effectiveness in capturing the users's correct intent.
- If the WER is high, the main objective semantic communication) fails.

Delivery Reliability

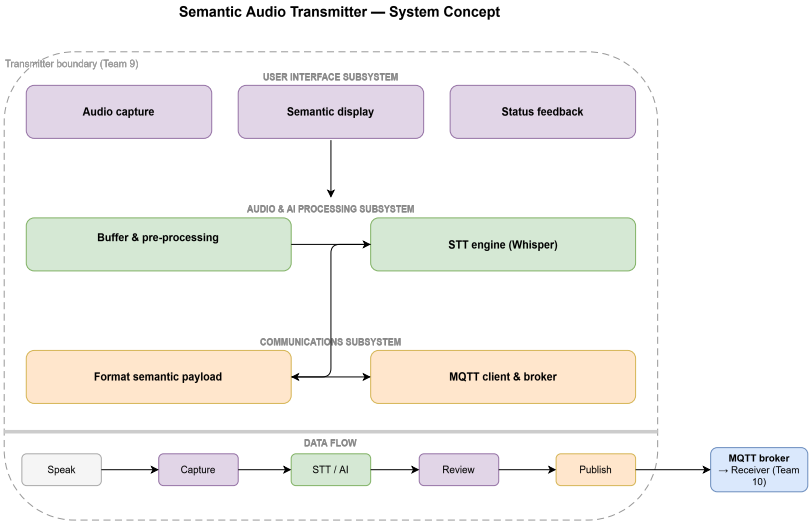
- The percentage of payload that are successfully published to the Broker out of the total number of attempts.
- The percentage of payloads that are compliant with the schema chosen to format the payload (JSON).

End-to-End Latency

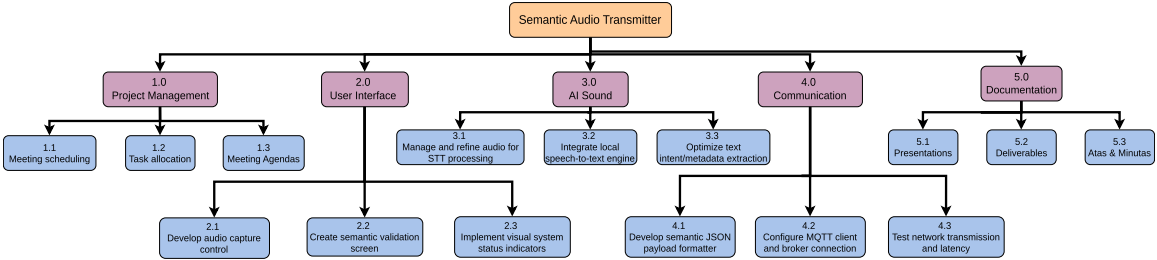
[MQTT Delivery Reliability]

- The total time measured from the moment the user finishes speaking until the formatted payload is sent.
- The risk report stipulates that the model's latency cannot exceed the defined limit.

System Concept



Work Breakdown Structure + Gantt

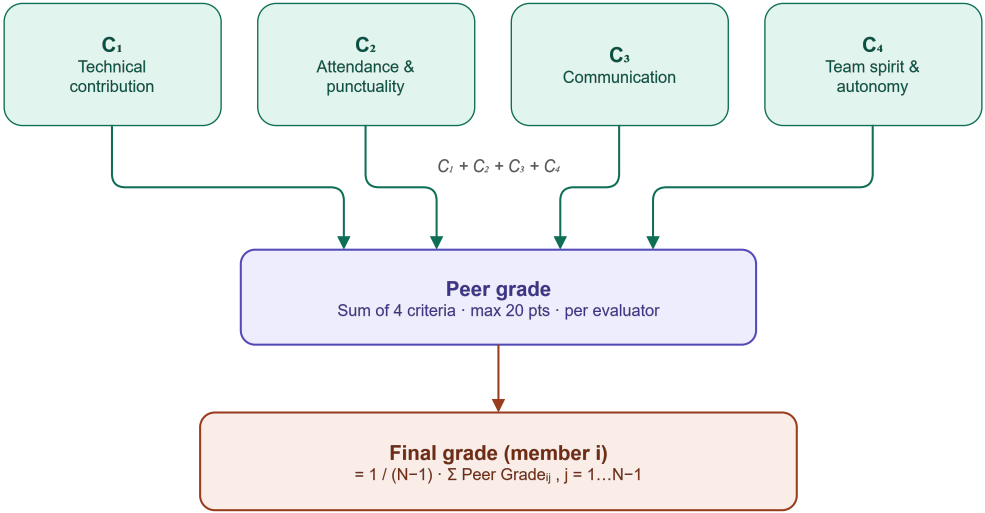


Risk Management

PoO \ Impact	Low	Medium	High
High			High Latency From AI Model
Medium		Insufficient Audio Quality	MQTT Communications Failures
Low		Resource Exhaustion During Demos	HW Failures

PoO - Probability of Occurence

Internal Evaluation



End of Semester Plan

- The next step is to move from preliminary design into concrete implementation of the user interface, the audio-conditioning chain, the semantic processing module, the payload builder, and the MQTT client [8].
- In parallel, the team needs to consolidate the transmitter-receiver interface, keep architectural consistency with previous deliverables, and continue integration testing under realistic local-demo conditions [9, 8].
- The end-of-semester objective is a stable demonstrator that preserves the validated requirements, supports local operation, and is ready for evaluation and live presentation [5, 7, 8].

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[9] Team 9.
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